QUANTIFYING SEMANTIC ALIGNMENT ACROSS LANGUAGES

WHICH TRANSLATION EQUIVALENTS ACTUALLY MEAN THE SAME THING?

BILL THOMPSON
biltho@mpi.nl
MPI for Psycholinguistics

SEAN ROBERTS
sean.roberts@bristol.ac.uk
University of Bristol

GARY LUPYAN
lupyan@wisc.edu
University of Wisconsin, Madison

Motivation

Do all languages convey semantic knowledge in the same way? To what extent do natural language semantics **mirror versus impose structure on the natural world**?

Q1: Which semantic domains are most and least alignable across languages?

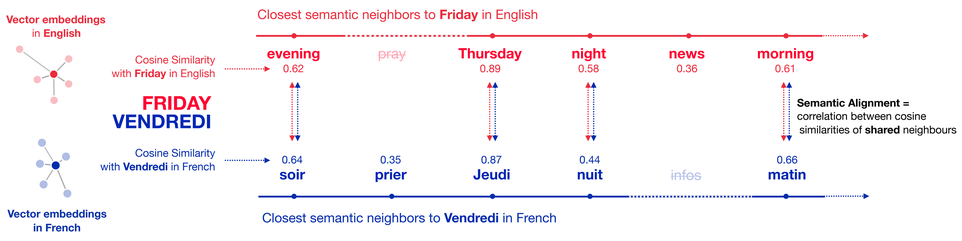
Q2: Do closely related languages show closely related semantic structures?

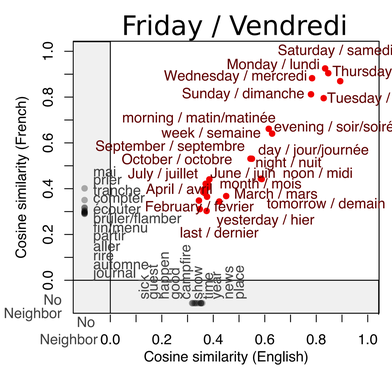
Q3: What is the relationship between semantics and culture?

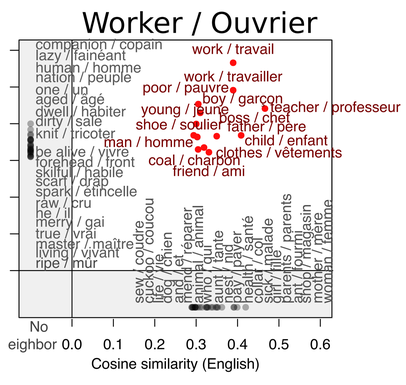
Methods

We combined cross-linguistic **vector embeddings, parallel translations, concept sets, a large dataset of cultural variation, and phylogenetic trees** to examine ***semantic alignment***: the structural similarity of a word's semantic network in **1485 unique language pairs, for 1000 concepts, across 18 semantic domains.**

Methods: *Computing Alignment*







Methods: *Resources*

Semantic Vectors, 150+ Langs.
fasttext.cc

1k Concepts, 107 Langs.
northeuralex.org

1400 Human Cultures
d-place.org

Semantic Domains:
ids.clld.org

Phylogenetic Tree:

Mapping the Origins and Expansion of the Indo-European Language Family

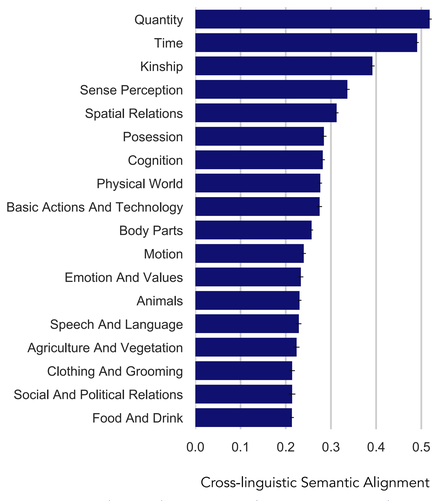
Resource Intersection: ***54 Languages***

Abkhazian	Chuvash	Hungarian	Modern Greek	Tamil
Albanian	Croatian	Icelandic	Northern Sami	Tatar
Armenian	Czech	Irish	Norwegian	Telugu
Avaric	Danish	Italian	Ossetian	Turkish
Bashkir	Dutch	Japanese	Polish	Ukrainian
Basque	English	Kalaallisut	Portugues	Welsh
Belarusian	Finnish	Kannada	Romanian	
Bengali	French	Kazakh	Russian	
Breton	Georgian	Korean	Slovak	
Bulgarian	German	Latvian	Slovenian	
Catalan	Hebrew	Lithuanian	Spanish	
Chechen	Hindi	Malayala	Swedish	

Results

Q1: Which semantic domains are most and least alignable across languages?

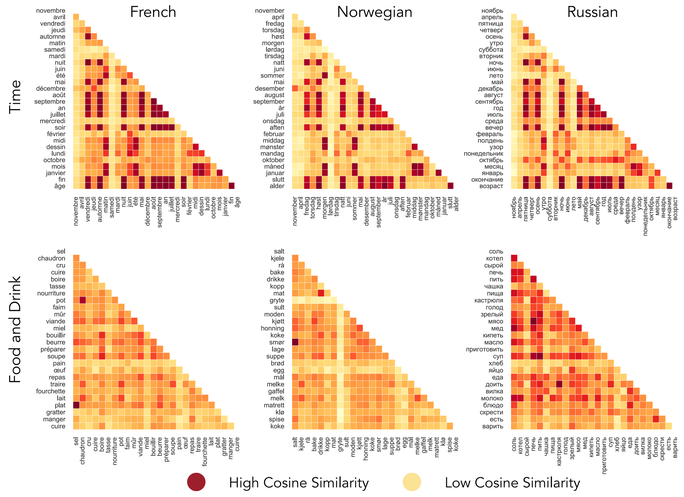
Words relating to e.g. **Quantity** and **Time align best; Basic Actions, Emotions** in the **middle Social/Political Relations, Food** the **least**



*our domain data are currently noisy — we're working on this!

Results (Cont.)

Q1: Which semantic domains are most and least alignable across languages?

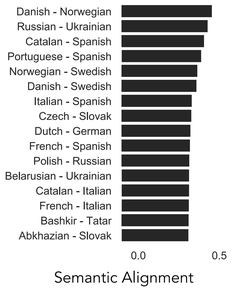


LOW ALIGNMENT EXAMPLES:
well :: bien, current :: courant, make :: faire
иметь :: own, есть :: eat, за :: behind

HIGH ALIGNMENT EXAMPLES:
play :: jouer, week :: semaine, friend :: ami
часто :: often, игра :: game, или :: or

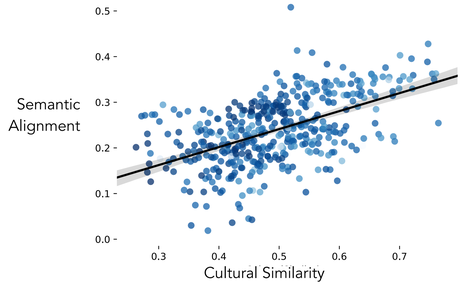
Q2: Do closely related languages show closely related semantic structures?

More **phylogenetically related** languages show **greater semantic alignment**.



Q3: What is the relationship between semantics and culture?

Languages spoken by **more similar cultures** show **greater alignment**



Conclusions

We found large-scale evidence for **language-specific structures in lexical semantics**, and for **a relationship between natural language semantics, phylogeny, and culture (evidence of linguistic adaptation?).** Even seemingly objective domains (e.g. *Animals, Body Parts*) exhibit considerable cross-linguistic variation. Words carve up the world in human-constructed ways.

References:
- P. Bojanowski*, E. Grave*, A. Joulin, T. Mikolov, Enriching Word Vectors with Subword Information. arXiv:1607.04606
- Bouckaert, Remco, et al. (2012) "Mapping the origins and expansion of the Indo-European language family." *Science*
- Kirby, Kathryn R., et al. (2016) "D-PLACE: A global database of cultural, linguistic and environmental *diversity." PloS one*
- Johannes Dellert and Gerhard Jäger (eds.). 2017. NorthEuraLex (version 0.9)
- Key, Mary Ritchie & Comrie, Bernard (eds.) 2015. The Intercontinental Dictionary Series.

Download the poster →